

Lecture Notes

EC226 Term 2 2025/26

Introduction to Time Series Models

Reading:

- Stock and Watson (2003): Chapter 14
- Dougherty (2016): Chapter 11 + 12
- Wooldridge (2013): Chapter 14 + 15

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1 Introduction

To this point we have worked with models of the following form:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i, \text{ for } i = 1, 2 \dots n \quad (1)$$

whereas we are now looking at models of the form:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \dots + \beta_k x_{kt} + \varepsilon_t, \text{ for } t = 1990Q1, 1990Q2 \dots \quad (2)$$

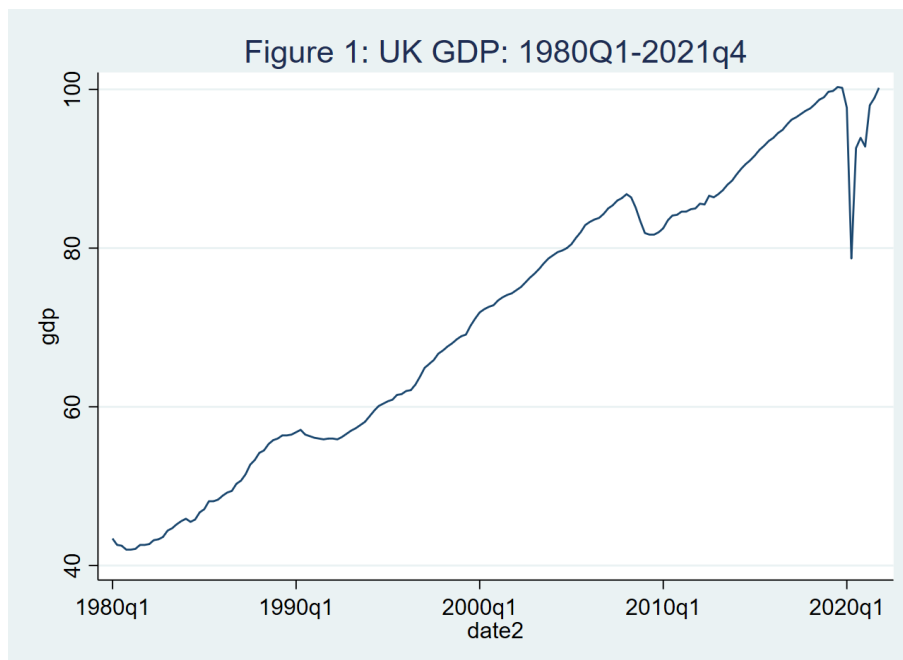
This is more than a change for a subscript, i , to a subscript, t . Time series data (data collected on a unit of observation over time) may exhibit a number of different features and these are often broken down into three components:

- trend component
- seasonal component
- cycle component

:

1.1 Trend

This refers to the increase or decrease in the series over time (see Figure 1).



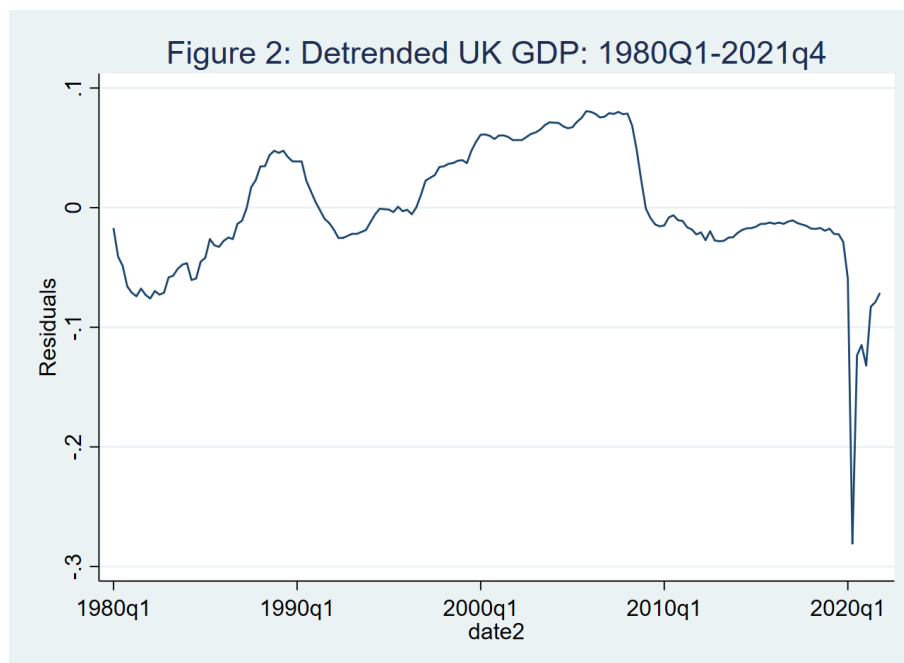
To get a feel for the extent of the time trend if we take $\lgdp = \ln(\text{gdp})$ and regress this on a variable which starts at 1 and increases by 1 for each new data point, then one gets:

```
. regress lgdp time
```

Source	SS	df	MS	Number of obs	=	169
Model	11.4480738	1	11.4480738	F(1, 167)	=	4305.01
Residual	.444093974	167	.002659245	Prob > F	=	0.0000
Total	11.8921678	168	.070786713	R-squared	=	0.9627
				Adj R-squared	=	0.9624
				Root MSE	=	.05157

lgdp	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
time	.005335	.0000813	65.61	0.000	.0051745	.0054955
_cons	3.776826	.0079689	473.95	0.000	3.761093	3.792558

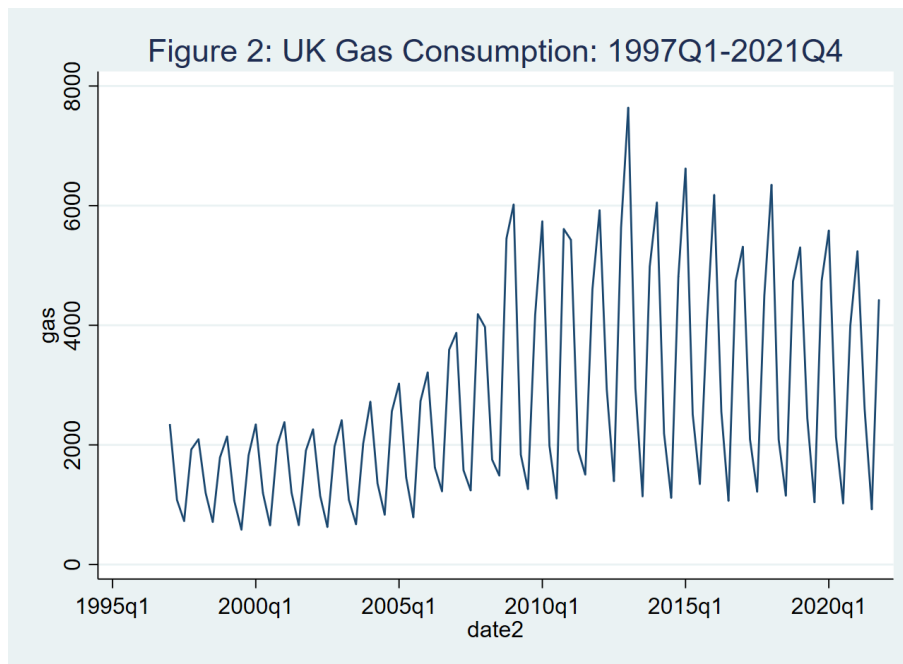
And we see from the coefficient on the trend that GDP in the UK since 1980 has grown on average by approximately 0.5% per quarter. If we take the residuals from the above regression and plot this, we have a detrended series of log(real GDP) as:



The residuals show no clear trend, but there is still a lot of information in the series, in particular about the cycle (see section 1.4) in GDP, but also the extent to which the UK economy shrank as a result of Covid-19 and the extent to which the UK economy remains below the long-run growth rate following the 2008 great recession.

1.2 Seasonal

This refers to the fluctuations in the data with a regular period e.g. retail sales are generally higher in quarter 4 (see Figure 2 below, which contain information on gas expenditure in the UK since 1997).



The plot clearly exhibits a trend, but it also exhibits a strong quarterly pattern. If we look at the actual data we have:

date	gas
1997q1	2350
1997q2	1080
1997q3	728
1997q4	1924
1998q1	2095
1998q2	1198
1998q3	712
1998q4	1789
1999q1	2140
1999q2	1066
1999q3	583
1999q4	1828

With gas consumption increasing from Q2 to Q3 to Q4 and peaking in Q1 of each year. Imagine treating the quarter of the year as a categorical variable with 4 outcomes: Quarter 1, Quarter 2, Quarter 3 and Quarter 4. Then one can construct 3 dummy variables (taking, e.g. Q3 as the default) and run the regression for $\ln(\text{gas})$:

```
. reg lgas ib3.quart if(date2<=tq(2021Q4))
```

Source	SS	df	MS	Number of obs	=	100
Model	31.9571256	3	10.6523752	F(3, 96)	=	75.41
Residual	13.5611039	96	.141261499	Prob > F	=	0.0000
				R-squared	=	0.7021
				Adj R-squared	=	0.6928
Total	45.5182294	99	.459780095	Root MSE	=	.37585

lgas	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
quart						
1	1.415792	.1063058	13.32	0.000	1.204777	1.626807
2	.5760641	.1063058	5.42	0.000	.3650488	.7870794
4	1.25777	.1063058	11.83	0.000	1.046755	1.468785
_cons	6.88775	.0751695	91.63	0.000	6.738539	7.03696

One can see e.g. that on average Q2 consumption is bigger than Q3 consumption by approximately 78% ($100 \times \exp(0.576) - 1$), Q4 consumption is above Q3 consumption by approximately 252% ($100 \times \exp(1.258) - 1$), and Q1 consumption is above Q3 consumption by 320% ($100 \times \exp(1.416) - 1$).

1.3 Cycle

The cyclical component is the part of the movement in the variable which can be explained by other cyclical movements in the economy and is sometimes referred to as the long-period effect or boom-bust process.

For example, during recessions, business cycles are usually characterized by slower growth rates than before the recession started (see the detrended lgdp in Figure 2).

2 Lags and interpreting coefficients

This refers to a delay in the response between x and y as macroeconomic variables are often sluggish to adjust to changes. This contrasts with cross-sectional data where it is reasonable to assume that the choice of individual, i , does not affect the choice of some other randomly sampled individual, j . However, for time series data it may well be the case that information from period $(t - 1)$ affects outcomes in period t .

Because of this sluggishness we will often consider dynamic models, which are a generalisation of the model specified in (2). In particular, we write the general dynamic model as:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \beta_3 x_{1t-1} + \beta_4 x_{1t-2} + \beta_5 y_{t-1} + \varepsilon_t \quad (3)$$

The issue with these models is the interpretation of the coefficients. In particular what is the effect on the variable y of a unit increase in x_1 .

If we (partially) differentiate y_t with respect to x_{1t} , we get:

$$\beta_1 = \left. \frac{dy_t}{dx_{1t}} \right|_{\text{Holding all else constant}}$$

However,

$$\beta_3 \neq \left. \frac{dy_t}{dx_{1t-1}} \right|_{\text{Holding all else constant}}$$

as it is not possible to change x_{1t-1} and hold y_{t-1} constant (as y_{t-1} depends on x_{1t-1}), as can be seen from equation (1) lagged 1 period!

Consider, the partial derivative of equation (3) with respect to (w.r.t.) x_{1t} , we get:

$$\frac{\partial y_t}{\partial x_{1t}} = \beta_1 \underbrace{\frac{\partial x_{1t}}{\partial x_{1t}}}_1 + \beta_2 \underbrace{\frac{\partial x_{2t}}{\partial x_{1t}}}_0 + \beta_3 \underbrace{\frac{\partial x_{1t-1}}{\partial x_{1t}}}_0 + \beta_4 \underbrace{\frac{\partial x_{1t-2}}{\partial x_{1t}}}_0 + \beta_5 \underbrace{\frac{\partial y_{t-1}}{\partial x_{1t}}}_0 \quad (4)$$

hence,

$$\frac{\partial y_t}{\partial x_{1t}} = \beta_1$$

Now consider equation (3) for y_{t+1}

$$y_{t+1} = \beta_0 + \beta_1 x_{1t+1} + \beta_2 x_{2t+1} + \beta_3 x_{1t} + \beta_4 x_{1t-1} + \beta_5 y_t + \varepsilon_{t+1} \quad (5)$$

and we partially differentiate equation (5) w.r.t. x_{1t} :

$$\frac{\partial y_{t+1}}{\partial x_{1t}} = \beta_1 \underbrace{\frac{\partial x_{1t+1}}{\partial x_{1t}}}_0 + \beta_2 \underbrace{\frac{\partial x_{2t+1}}{\partial x_{1t}}}_0 + \beta_3 \underbrace{\frac{\partial x_{1t}}{\partial x_{1t}}}_1 + \beta_4 \underbrace{\frac{\partial x_{1t-1}}{\partial x_{1t}}}_0 + \beta_5 \underbrace{\frac{\partial y_t}{\partial x_{1t}}}_{\beta_1} \quad (6)$$

in which case:

$$\frac{\partial y_{t+1}}{\partial x_{1t}} = \beta_3 + \beta_5\beta_1$$

Now consider equation (3) for y_{t+2} :

$$y_{t+2} = \beta_0 + \beta_1 x_{1t+2} + \beta_2 x_{2t+2} + \beta_3 x_{1t+1} + \beta_4 x_{1t} + \beta_5 y_{t+1} + \varepsilon_{t+2} \quad (7)$$

and we partially differentiate equation (7) w.r.t. x_{1t}

$$\frac{\partial y_{t+2}}{\partial x_{1t}} = \beta_1 \underbrace{\frac{\partial x_{1t+2}}{\partial x_{1t}}}_0 + \beta_2 \underbrace{\frac{\partial x_{2t+2}}{\partial x_{1t}}}_0 + \beta_3 \underbrace{\frac{\partial x_{1t+1}}{\partial x_{1t}}}_0 + \beta_4 \underbrace{\frac{\partial x_{1t}}{\partial x_{1t}}}_1 + \beta_5 \underbrace{\frac{\partial y_{t+1}}{\partial x_{1t}}}_{\beta_3 + \beta_5\beta_1} \quad (8)$$

then

$$\frac{\partial y_{t+2}}{\partial x_{1t}} = \beta_4 + \beta_5(\beta_3 + \beta_5\beta_1)$$

Now consider equation (3) for y_{t+3} :

$$y_{t+3} = \beta_0 + \beta_1 x_{1t+3} + \beta_2 x_{2t+3} + \beta_3 x_{1t+2} + \beta_4 x_{1t+1} + \beta_5 y_{t+2} + \varepsilon_{t+3} \quad (9)$$

and we partially differentiate equation (9) w.r.t. x_{1t}

$$\frac{\partial y_{t+3}}{\partial x_{1t}} = \beta_1 \underbrace{\frac{\partial x_{1t+3}}{\partial x_{1t}}}_0 + \beta_2 \underbrace{\frac{\partial x_{2t+3}}{\partial x_{1t}}}_0 + \beta_3 \underbrace{\frac{\partial x_{1t+2}}{\partial x_{1t}}}_0 + \beta_4 \underbrace{\frac{\partial x_{1t+1}}{\partial x_{1t}}}_0 + \beta_5 \underbrace{\frac{\partial y_{t+2}}{\partial x_{1t}}}_{\beta_4 + \beta_5(\beta_3 + \beta_5\beta_1)} \quad (10)$$

then

$$\frac{\partial y_{t+3}}{\partial x_{1t}} = \beta_5 [\beta_4 + \beta_5(\beta_3 + \beta_5\beta_1)]$$

If we think of accumulating these responses, we get:

0 Period (contemporaneous) total change is:

$$\frac{\partial y_t}{\partial x_{1t}} = \beta_1$$

1 period total change is:

$$\frac{\partial y_t}{\partial x_{1t}} + \frac{\partial y_{t+1}}{\partial x_{1t}} = \beta_1 + \beta_3 + \beta_5\beta_1$$

2 period total change is:

$$\frac{\partial y_t}{\partial x_{1t}} + \frac{\partial y_{t+1}}{\partial x_{1t}} + \frac{\partial y_{t+2}}{\partial x_{1t}} = \beta_1 + \beta_3 + \beta_4 + \beta_5(\beta_1 + \beta_3) + \beta_5^2\beta_1$$

3 period total change is:

$$\frac{\partial y_t}{\partial x_{1t}} + \frac{\partial y_{t+1}}{\partial x_{1t}} + \frac{\partial y_{t+2}}{\partial x_{1t}} + \frac{\partial y_{t+3}}{\partial x_{1t}} = \beta_1 + \beta_3 + \beta_4 + \beta_5(\beta_1 + \beta_3 + \beta_4) + \beta_5^2(\beta_1 + \beta_3) + \beta_5^3\beta_1$$

In the long-run as $x_{1t} = x_{1t-1} = \dots = x_{1t-s} = x_1^*$, $x_{2t} = x_{2t-1} = \dots = x_{2t-s} = x_2^*$, $y_t = y_{t-1} = \dots = y_{t-s} = y^*$ then we have from equation (3)

$$y^* = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \beta_3 x_1^* + \beta_4 x_1^* + \beta_5 y^* + \varepsilon^*$$

collecting the same variables together, we get:

$$y^*(1 - \beta_5) = \beta_0 + (\beta_1 + \beta_3 + \beta_4)x_1^* + \beta_2 x_2^* + \varepsilon^*$$

and solving for y^*

$$y^* = \frac{\beta_0}{(1 - \beta_5)} + \frac{(\beta_1 + \beta_3 + \beta_4)}{(1 - \beta_5)}x_1^* + \frac{\beta_2}{(1 - \beta_5)}x_2^* \quad (11)$$

In summary then we have,

Lags	Multiplier
0	β_1
1	$\beta_1 + \beta_3 + \beta_5\beta_1$
2	$\beta_1 + \beta_3 + \beta_4 + \beta_5(\beta_1 + \beta_3) + \beta_5^2\beta_1$
3	$\beta_1 + \beta_3 + \beta_4 + \beta_5(\beta_1 + \beta_3 + \beta_4) + \beta_5^2(\beta_1 + \beta_3) + \beta_5^3\beta_1$
4	$\beta_1 + \beta_3 + \beta_4 + \beta_5(\beta_1 + \beta_3 + \beta_4) + \beta_5^2(\beta_1 + \beta_3 + \beta_4) + \beta_5^3(\beta_1 + \beta_3) + \beta_5^4\beta_1$
5	$\beta_1 + \beta_3 + \beta_4 + \beta_5(\beta_1 + \beta_3 + \beta_4) + \beta_5^2(\beta_1 + \beta_3 + \beta_4) + \beta_5^3(\beta_1 + \beta_3 + \beta_4) + \beta_5^4(\beta_1 + \beta_3) + \beta_5^5\beta_1$
...	...
∞	$\frac{\beta_1 + \beta_3 + \beta_4}{1 - \beta_5}$

3 Lagged dependent variable

Take a simple dynamic model:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \varepsilon_t \quad t = 1, \dots, T \quad (12)$$

and let us consider the assumptions we make in this case compared to the CLRM assumptions we used in Term 1. In Term 1 we assumed:

1. $E(\varepsilon_i | x_1, \dots, x_n) = E(\varepsilon_i) = 0$ (so the error term has mean zero, independent of x_i 's).
2. $V(\varepsilon_i | x_1, \dots, x_n) = \sigma^2$ (error variance is constant (homoscedastic) – points are distributed around the true regression line with a constant spread)
3. $cov(\varepsilon_t, \varepsilon_{t-h}) = 0$ (the errors are serially uncorrelated over observations)
4. $\varepsilon_i | x_1, \dots, x_n \sim N(0, \sigma^2)$

Assumption 1 is a strict exogeneity assumption and implies that the errors of individual i are not correlated with the values of the regressors for individual i and for all other individuals in the sample, i.e. $E(\varepsilon_i | x_1, x_2, \dots, x_n) = 0$.

But in terms of equation (12) this would imply: $E(\varepsilon_t | y_1, y_2, \dots, y_T) = 0$ and that is clearly not sensible as from (12) $cov(y_t, \varepsilon_t) \neq 0$ and as this condition is needed for unbiasedness, then OLS estimation of (12) yields a biased coefficient estimator (see Appendix A, which shows the distribution of the OLS estimator of b_1 and how $E(b_1) \rightarrow \beta_1$ only for large sample sizes).

Consequently we replace assumption (1) with a weaker condition which is the condition of contemporaneous exogeneity, i.e. $E(\varepsilon_t | y_1, y_2, \dots, y_{t-1}) = 0$, which implies that the errors are uncorrelated with past values of the dependent variable.

In addition, we need to replace the assumption of random sampling in cross-sectional data (which implies the decisions/outcomes of one person, firm, country (observation) do not impact on the outcomes of another observation. In time series as observations are related over time, then a change in y in 1980 is likely to have an effect on this variable in 1981. In effect this means we have a single sequence of data. To address this we replace the assumption of Random Sampling with two conditions:

1. Stationarity
2. Weak Dependence

3.1 Stationarity

This requires that a sequence $\{y_t\}$ satisfies:

1. $E(y_t) = \mu$ (mean is constant at all points in time)

2. $V(y_t) = \sigma_y^2$ (variance is constant at all points in time)
3. $cov(y_t, y_{t-h}) = \gamma_h$ (covariance only depend on h and not t)

3.2 Weak Dependence

This requires:

$$cov(y_t, y_{t-h}) \rightarrow 0 \text{ as } h \rightarrow \infty.$$

We will return to these ideas in the next two handouts.

In general, if we consider any dynamic model, like:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 x_t + \varepsilon_t \quad t = 1, \dots, T$$

Then if we assume:

$$E(\varepsilon_t | y_1, y_2, \dots, y_{t-1}, x_t) = 0$$

and given the condition of stationarity and weak dependence:

1. Our OLS estimator is not unbiased, but is consistent (implying that as the number of observations, T , gets large our OLS estimator gets “close” to the true coefficient) (see Appendix A for a demonstration of this idea).
2. The distribution of the OLS estimator is asymptotically normally distributed (implying that as the number of observations, T , gets large the distribution of the OLS estimator becomes like a normal distribution). This implies for hypothesis testing of a single restriction you should use the normal distribution NOT the t-distribution and for testing multiple restrictions you should use the χ^2 -distribution rather than the F-distribution (see Appendix A for a demonstration of this idea).

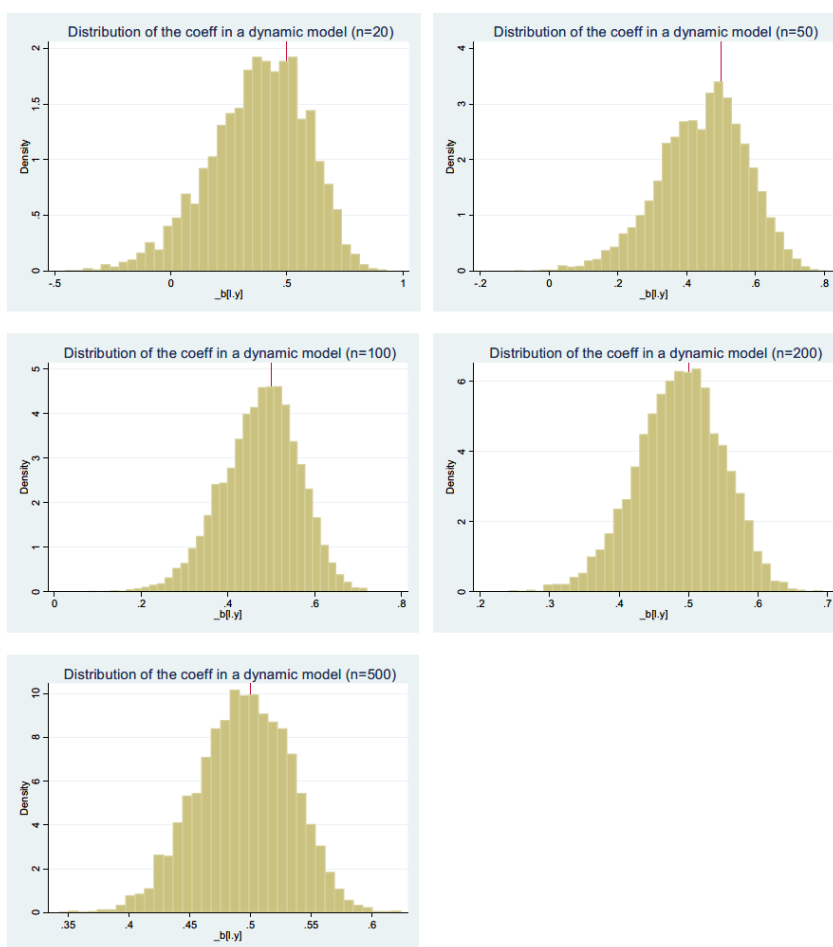
Stata (R) commands one might use to specify the data as time series and to run a regression are briefly covered in Appendix (D).

Appendices

A Biases in dynamic regression models - Stata

The model:

$$y_t = 0.5y_{t-1} + \varepsilon_t, \quad t = 1, \dots, T \text{ and } \varepsilon_t \sim N(0, 1)$$



Variable	Obs	Mean	Std. Dev.	Min	Max
b_ly	5000	.376677	.2071881	-.4529505	.9300346
b_ly	5000	.447168	.1267887	-.0990768	.8080268
b_ly	5000	.4755448	.0878672	.0767301	.7217281
b_ly	5000	.4869785	.0622082	.2412497	.6938383
b_ly	5000	.4950009	.0385217	.3425669	.6241989

and the associated Stata commands to generate these graphs:

```
clear
/* This empties Stata's memory of any previous programs */
prog drop _all
/* This sets the start point for random number generator */
set seed 1342
/* The sample size is 30 */
set obs 550
gen index=_n
tsset index
scalar alpha=0
scalar beta=0.5
/* This is the actual program where a new y is generated as alpha+beta*x+N(0,1) - we estimate
   model and save the coefficient estimates as _b_x and _b_a and the standard errors as se_a
   and the standard error of the regression as se_e */
   program define reg_sim, rclass
       gen y=0
       replace y = alpha + beta*y[_n-1] + rnormal() if(index>1)
       reg y l.y if(index>50 & index<=550)
return scalar b_ly = _b[l.y]
drop y
end

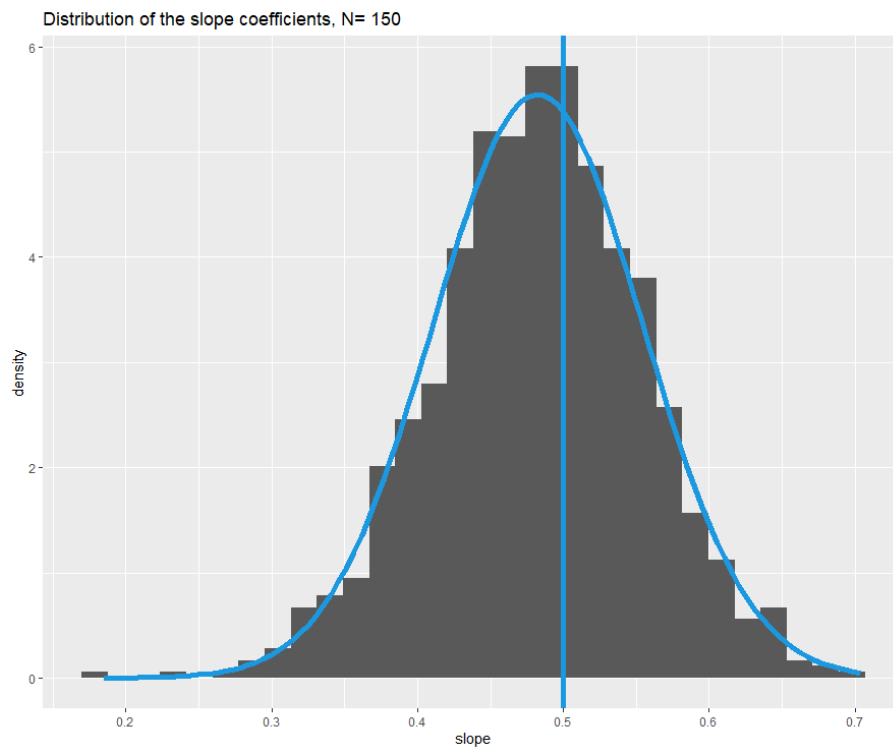
/* This tells Stata to do the regressions 5000 times and what values to save*/
simulate b_ly =_b[l.y], reps(5000): reg_sim

/* generates summary statistics */
su b_ly, detail
histogram b_ly, xline(0.5) title(Distribution of the coeff in a dynamic model (n=20))
```

B Biases in dynamic regression models - R

The model:

$$y_t = 0.5y_{t-1} + \varepsilon_t, \quad t = 1, \dots, T \text{ and } \varepsilon_t \sim N(0, 1)$$



and the associated R commands to generate these graphs:

```
set.seed(seed = 1342)
# The sample size is 30
n <- 150
alpha <- 0
beta <- 0.5
nrep <- 1000
b_1 <- numeric(nrep)

for (j in 1:nrep) {
  y <- 0
  e <- rnorm(n, mean=0, sd=1)
  for(i in 2:length(e)){
    y[i] <- alpha + beta * y[i-1] + e[i]
  }
  ly=lag(y)
  x <- data.frame(y, ly)
```

```
ols_fit <- lm(y ~ ly, data = x)
b_1[j] <- as.vector(ols_fit$coef[2])
df_coef <- data.frame(slope= b_1)
}

df_coef %>%
  summarise(m_slope = mean(slope),
            sd_slope = sd(slope))

ggplot(df_coef, aes(slope)) +
  geom_histogram(aes(y = ..density..)) +
  stat_function(fun = dnorm,
               args = list(mean = mean(df_coef$slope),
                           sd = sd(df_coef$slope)),
               col = "#1b98e0",
               size = 2) +
  geom_vline(xintercept = beta, color="#1b98e0", lwd =2) +
  ggtitle(paste("Distribution of the slope coefficients, N=", n))
```

C Stata Commands for time series models

To get Stata to understand the data is time series you need to declare the data as such.
So for annual data which is of the form:

	year	gdp
1.	1948	325962
2.	1949	336994
3.	1950	348137
4.	1951	361238
5.	1952	367027
6.	1953	387296
7.	1954	403966

Type:

```
tsset year
```

For quarterly data, which of the form:

	date	alc
1.	1985Q1	3176
2.	1985Q2	3381
3.	1985Q3	3471
4.	1985Q4	3800
5.	1986Q1	3270
6.	1986Q2	3576
7.	1986Q3	3674

Type:

```
generate time = quarterly(year, "yq")
```

```
format time %tq
```

```
tsset time
```

```
/*To generate lagged value you can use: */
```

```
gen lx=l.x
```

```
gen l2x=l2.x
```

```
/*To generate a first difference ( $\{x_t\} - \{x_{t-1}\}$ ) you can use: */
```

```
gen dx=d.x
```

```
/*To estimate a model of y on 1 lag of y, x and 1 lag of x by OLS one would type */
```

```
reg y l.y l(0/1).x
```


D R Commands for time series models

#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.

#I used the following libraries over Term 2:

```
library(tseries)
library(urca)
library(forecast)
library(ggplot2)
library(dplyr)
library(zoo)
library(xts)
library(ggfortify)
library(dyn)
library(lmtest)
library(car)
```

To get R to understand the data is time series you need to declare the data as such.
So for annual data which is of the form:

	date	gdp
1.	1948	325962
2.	1949	336994
3.	1950	348137
4.	1951	361238
5.	1952	367027
6.	1953	387296
7.	1954	403966
...	...	...

Type:

```
# This reads in the dataset from an excel file
uk_gdp_a <- read_excel("C:/Downloads/uk_gdp_a.xlsx")
# This drops the date series
uk_gdp_a <- uk_gdp_a %>%
  select(-c(date))
# This creates a new date variable
date2 <- seq(as.Date("1948-1-1"), as.Date("2014-1-1"), by = "years")
# This uses combines the gdp and the date2 series into a time series data
uk_gdp_tsa <- xts(uk_gdp_a, order.by=date2))
```

For quarterly data, which of the form:

	date	alc
1.	1985Q1	3176
2.	1985Q2	3381
3.	1985Q3	3471
4.	1985Q4	3800
5.	1986Q1	3270
6.	1986Q2	3576
7.	1986Q3	3674

Type:

```
# This reads in the dataset from an excel file
uk_alc_q <- read_excel("C:/Downloads/uk_alc_q.xlsx")
# This drops the date series
uk_alc_q <- uk_alc_q %>%
  select(-c(date))
# This creates the date series
date2 <- seq(as.Date("1955-1-1"), as.Date("2015-4-1"), by = "quarters")
# This uses xts to tell r this is time series data
uk_alc_tsq <- xts(uk_alc_q, order.by=date2))

# To generate lagged value you use:
uk_gdp_tsa$gdp_1 <- lag(uk_gdp_tsa$gdp,1)
uk_gdp_tsa$gdp_4 <- lag(uk_gdp_tsa$gdp,4)
# To generate a differenced series D.x_t:
uk_gdp_tsa$d2gdp <- diff(uk_gdp_tsa$gdp, lag = 1)
# To generate a differenced series D2.x_t=x_t-x_t-2:
uk_gdp_tsa$d2gdp <- diff(uk_gdp_tsa$gdp, lag = 2)
# To generate a differenced series D.D.x_t:
uk_gdp_tsa$d2gdp <- diff(uk_gdp_tsa$gdp, lag = 1)
# To generate a differenced series D.D.x_t:
uk_gdp_tsa$nd2gdp <- diff(diff(uk_gdp_tsa$gdp))

# To estimate a model of y on 1 lag of y, x and 1 lag of x by OLS one would type
dyn$lm(y ~ lag(y, 1) + x + lag(x, 1))
```

References

Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.

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Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.